DO NOW — RECALL

Lesson 097 · Little Man Computer: Branches and Loops

Name: _____ Class: _____ Date: _____

1. What does BRA do in LMC?

2. When does BRZ branch?

3. What does LDA COUNT do?

4. Stack Push2,Push5,Pop,Push8: final bottom-to-top?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Branches unconditionally to a label.

2. Model answer

When accumulator equals zero.

3. Model answer

Loads COUNT's mailbox contents into ACC.

4. Model answer

[2,8].

By the end, you can…

• BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

• The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

• Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

Vocabulary — know these terms

BRA — Always branch to label.

BRZ — Branch if ACC zero.

label — Symbolic instruction/memory address.

loop — Repeated control flow.

termination — Stop condition for loop.

NEW LEARNING

Read, annotate and explain

Observe

BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

Apply

The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

Evaluate

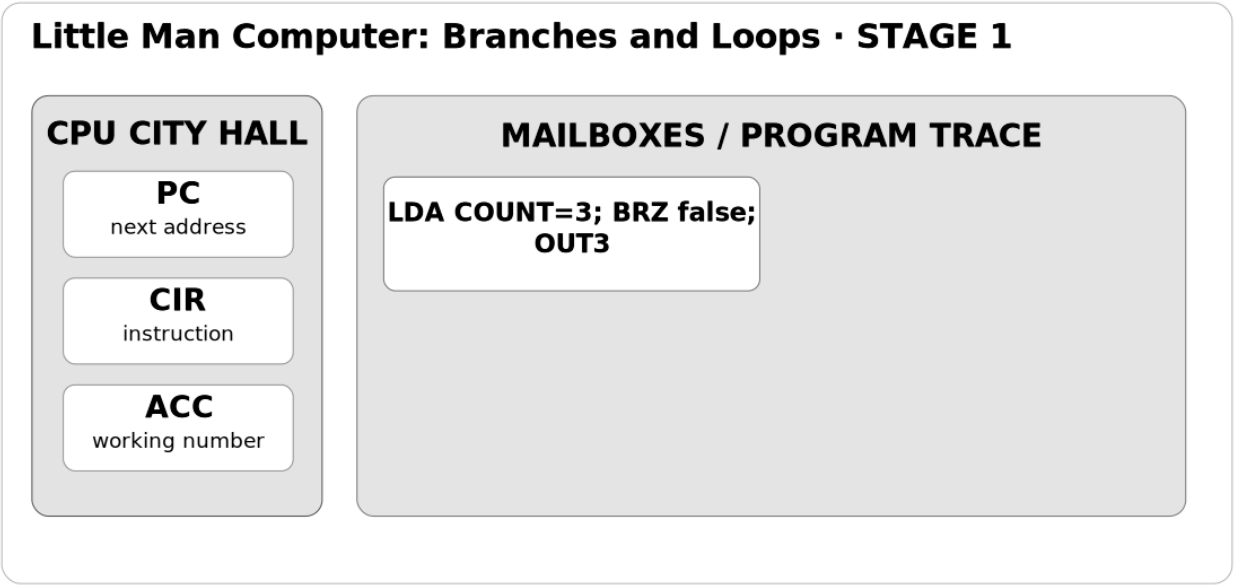
Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

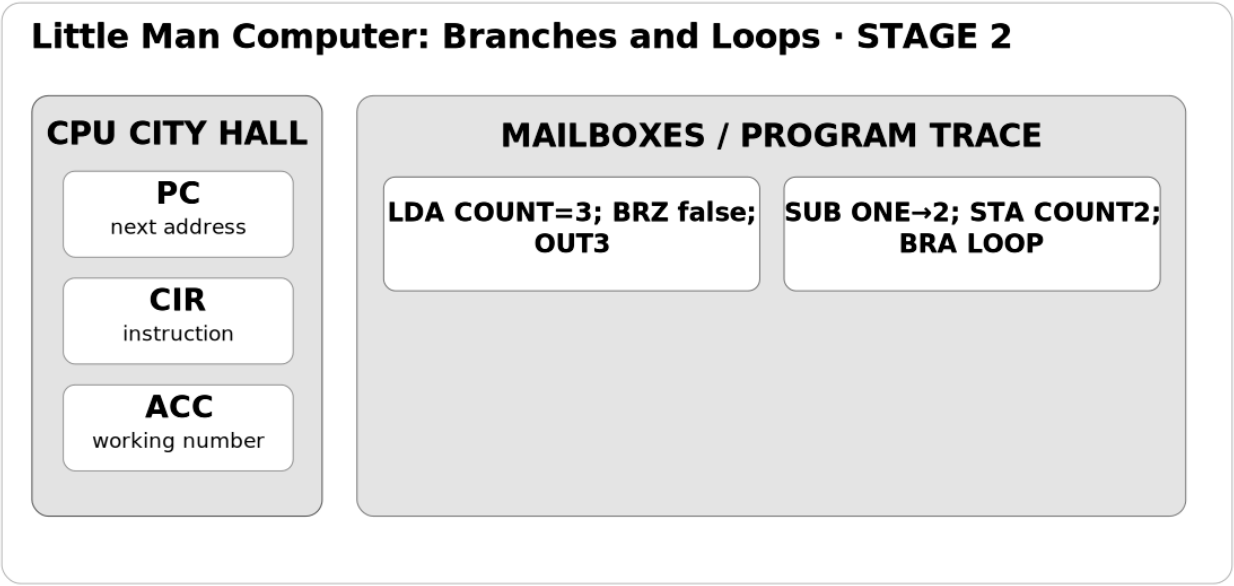
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer: Branches and Loops · STAGE 1; LDA COUNT=3; BRZ false; OUT3

New Learning

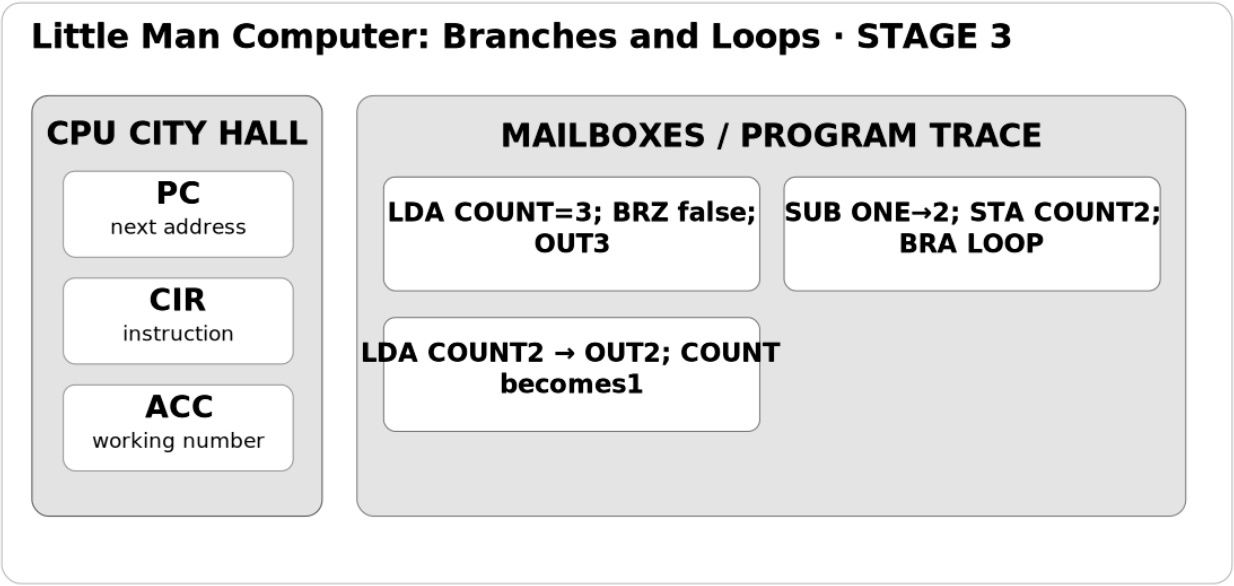


Little Man Computer: Branches and Loops · STAGE 2; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

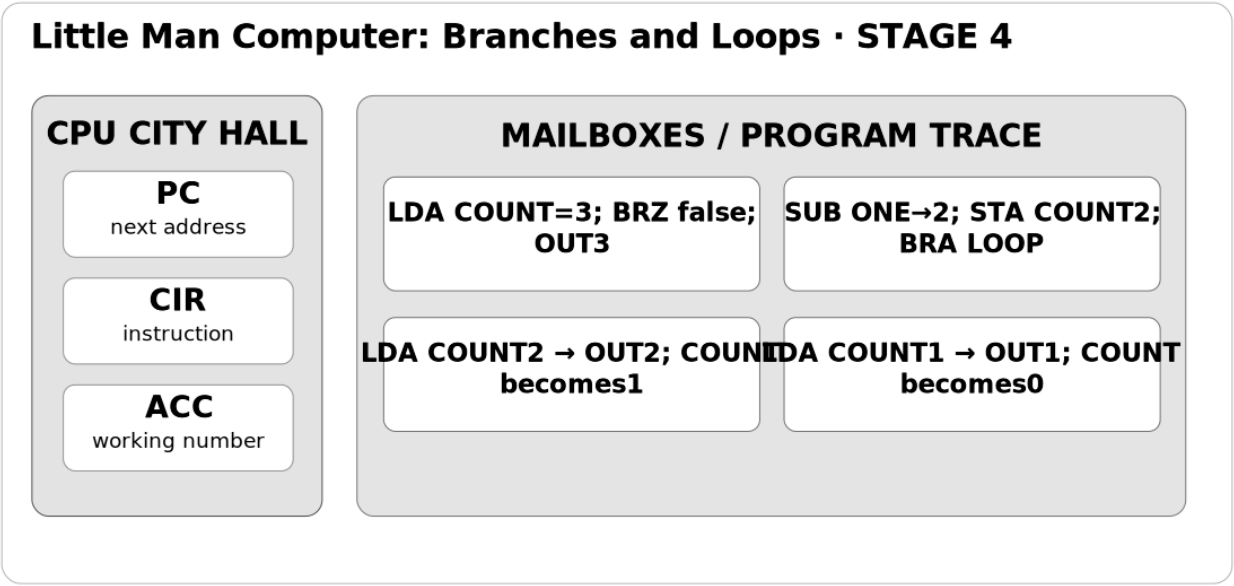


Little Man Computer: Branches and Loops · STAGE 3; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP; LDA COUNT2 → OUT2; COUNT becomes1

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

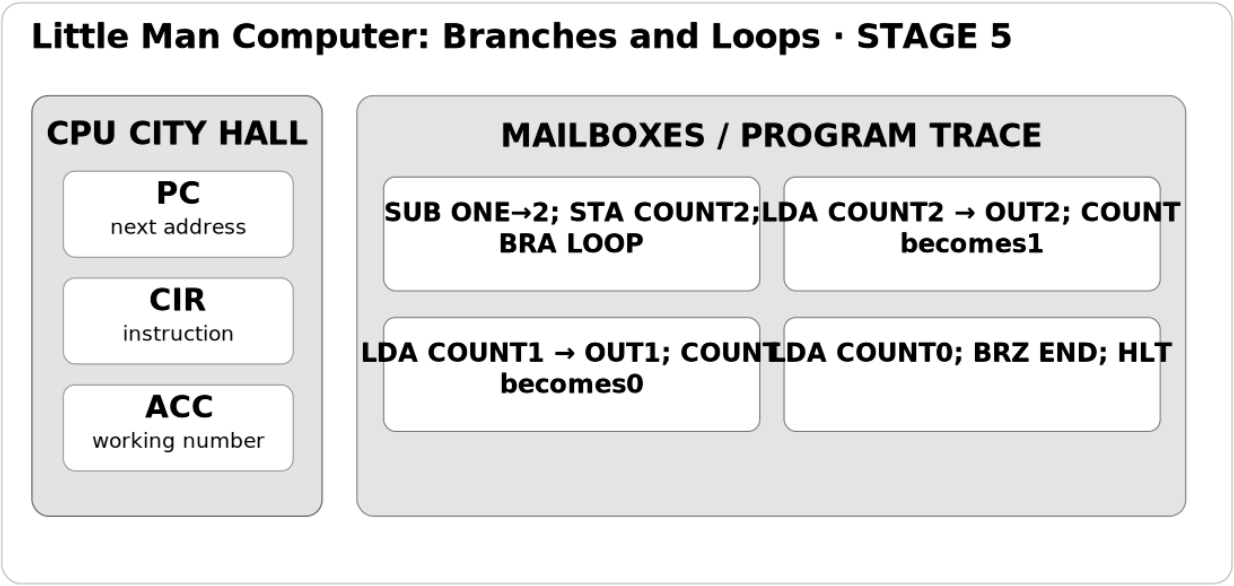


Little Man Computer: Branches and Loops · STAGE 4; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP; LDA COUNT2 → OUT2; COUNT becomes1; LDA COUNT1 → OUT1; COUNT becomes0

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer: Branches and Loops · STAGE 5; LDA COUNT=3; BRZ false; OUT3; SUB ONE→2; STA COUNT2; BRA LOOP; LDA COUNT2 → OUT2; COUNT becomes1; LDA COUNT1 → OUT1; COUNT becomes0; LDA COUNT0; BRZ END; HLT

I DO — WORKED EXAMPLE

Follow the worked method

Start COUNT=2 instead of3. What gets printed and why?

Follow the reasoning

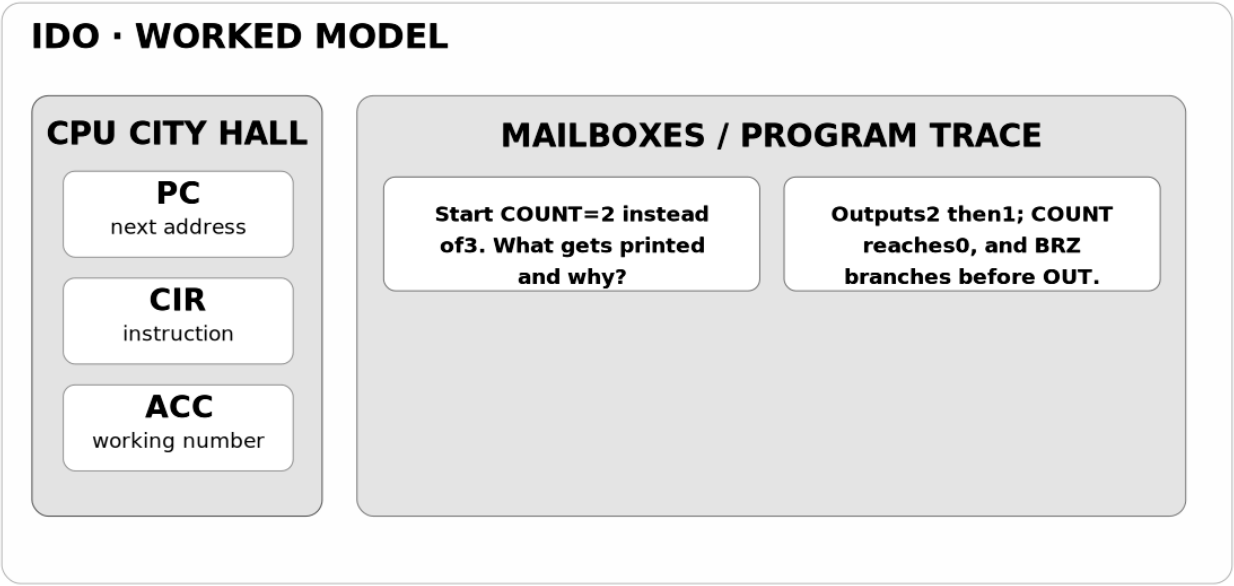
Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Start COUNT=2 instead of3. What gets printed and why?

Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.



IDO · WORKED MODEL; Start COUNT=2 instead of3. What gets printed and why?; Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace COUNT DAT3 loop outputs and COUNT after each SUB. [6]

Question 2

Change COUNT DAT3 to COUNT DAT2. State outputs and explain BRZ. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

A pupil replaces BRZ END with BRA END at LOOP. Explain effect and repair. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

BRZ END always jumps to END regardless of ACC.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

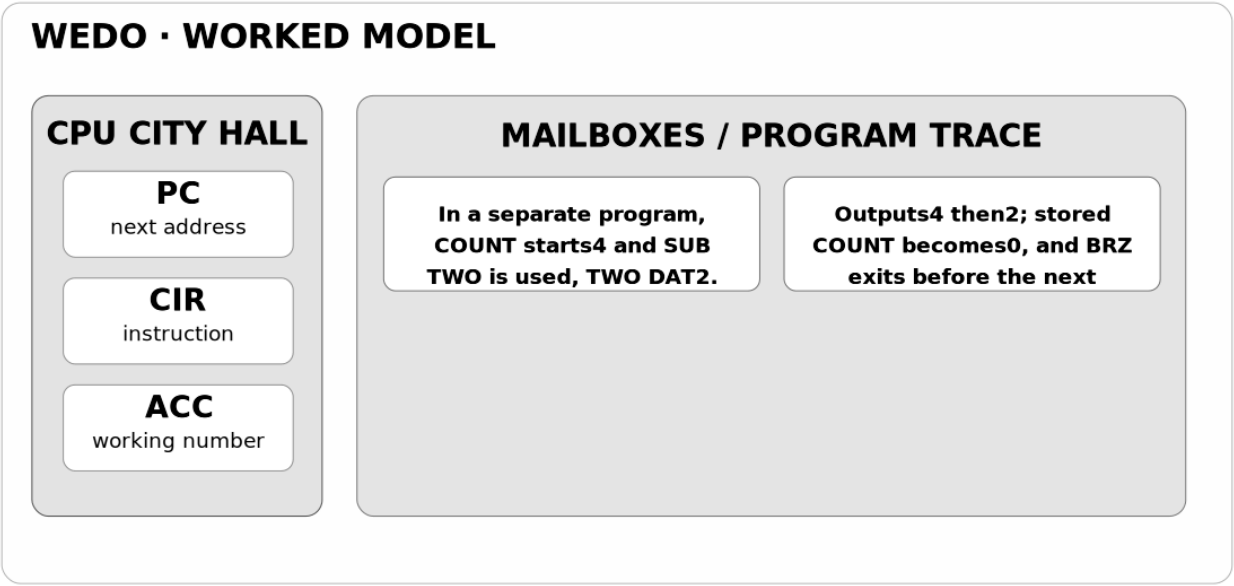
Follow the reasoning
Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.

Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.



WEDO · WORKED MODEL; In a separate program, COUNT starts4 and SUB TWO is used, TWO DAT2. Give outputs.; Outputs4 then2; stored COUNT becomes0, and BRZ exits before the next OUT.